

Karnataka Joint Gravity–Magnetic Inversion: Couplings, Remanence and Rock-Sample Constraints

The gravity and the magnetic data of the same area inverted together on one mesh with terrain, once for each coupling of the joint inversion, and the magnetic data alone with a magnetization vector per cell. The report shows what each coupling does to the two models, what it costs in data fit, which assumption suits this belt, what the magnetic data say about the direction of magnetization, and how the measured densities of the rocks, used as bounds, change the models.

AREA	DATA	MODELS
70 × 70 km, easting 641–711 km, northing 1634–1704 km (UTM 43N, EPSG:32643)	Complete Bouguer anomaly, 5,040 points, error 0.5 mGal; magnetic anomaly continued to 1 km above the ground, 5,041 points, error 5% + 10 nT; second-order trends removed	Density contrast (–0.2 to +0.5 g/cc) and susceptibility (0 to 1 SI) on 335,518 cells below the ground (1 km × 1 km × 250 m); each sparse with $\alpha_s = 1$ and depth weighting $\beta = 1$
RUNS	CONSTRAINTS	
6 joint inversions and one magnetization-vector inversion on AWS EC2 (ap-south-1, c5.9xlarge and c5.18xlarge), 22–124 minutes each, \$13.62 in total; the same couplings on a 2 km mesh locally	The density bounds from the measured rock samples (–0.15 to +0.35 g/cc) with $\beta = 0.5$ for the gravity model: 2 runs at full resolution (\$1.29) and 12 on the 2 km mesh (Section 5)	

Summary

- **The data do not rank the couplings.** Every run fits both datasets (χ^2/N 0.88–1.10), and there is no known model of the area to compare with. The conclusions are of three kinds: what each coupling does (certain: a property of the method), what the magnetic data show (the remanence), and which coupling suits this belt (a judgement from the geology).
- **Without a coupling the joint inversion returns the two single inversions** (cell-by-cell correlation 0.95 and 0.76); the two uncoupled models have little in common cell by cell (21% of the magnetic cells are dense or light, 32% of the susceptibility edges lie on a density edge).
- **What each coupling does** (Section 4.1): the cross-gradient makes the gradients parallel (0.44 → 0.07) mostly by keeping the edges apart (2% shared); the joint total variation lines the edges up (47%) and leaves the density model as it was (0.97); the linear correspondence lifts the dense body to 0.0–3.8 km (uncoupled 3.0–7.5 km) and puts 45% of the density model below the core; the group lasso pairs the magnetic cells with dense ones (81%, 21% uncoupled) and lifts the dense body part of the way, to 2.0–5.2 km. The couplings that tie the values move the dense body up because the magnetic data pin the magnetic rock near the surface far more firmly than the gravity data pin the dense rock at depth (Section 4.2).
- **The magnetic data ask for remanent magnetization.** With a magnetization vector per cell the RMS residual falls from 49 to 28 nT, and at the 92 stations south of the Sandur belt that every susceptibility model underfits by more than 150 nT from 321 to 129 nT (22 of them still off by more than 150 nT); the strongly magnetized cells point at I 74°, D -79°, not along the present field (I 19°, D -1°).
- **For this belt a structural coupling is the one consistent with the geology** (Section 4.3), a judgement from the rock samples and the mapping, not from the fit: the magnetic iron formation lies on the margins of the dense, weakly magnetic greenstone, so the two models share boundaries, not values. The linear correspondence's premise is contradicted by the samples; the group lasso's holds for the iron formation only. Section 4.4 gives the first choice for other settings.
- **No coupling fixes the depth.** Where the couplings disagree, as on the depth of the dense body, the data do not decide; the depth follows the assumption and the regularization.
- **The rock samples move the density model more than any coupling** (Section 5). Bounded by the measured densities (–0.15 to +0.35 g/cc) and with a depth weighting of $\beta = 0.5$ for the gravity model, the dense body reaches the ground where the belt crops out, and under the main high ends at 5.5 km (half-maximum 1.0–5.2 km, against 3.0–7.5 km), with the data fitted as well (χ^2/N 0.91–0.94). The constrained density model correlates at 0.42 with the unconstrained one; the coupled and uncoupled models of one setting at 0.98.

1 Set-up

The two datasets are those of the single-method reports: the Bouguer anomaly with the terrain correction at 1 km, and the magnetic anomaly continued upwards to 1 km above the ground. They share one mesh that follows the ground (84 × 84 × 50 cells, 335,518 below the ground), so that a cell of the density model and a cell of the susceptibility model are the same rock.

In the runs of the SimPEG couplings each model is regularized as in the reference run of its own report: sparse (IRLS) with $\alpha_s = 1$, $p = [0, 2, 2, 2]$ and Li & Oldenburg depth weighting with $\beta = 1$, within its bounds. One trade-off parameter serves both models; the pipeline balances the two regularizations

against their data and cools it until $\chi^2 = N$ for the 10,081 data together, with at most 60 iterations and 40 IRLS iterations. The coupling weight is 1 in the pipeline's unit-free scale (as strong as the regularization).

The group lasso has its own regularization: an L2 term ($\lambda_2 = 0.3$) and the group norm of each cell's (density, susceptibility) pair, solved by ADMM within the same bounds. Its cells are weighed like those of the other runs (cell volume $\times$ the depth weight with $\beta = 1$), each datum by its error, λ_1 is chosen on a sweep of 8 values for $\chi^2 = N$, and the two datasets are then reweighted until each fits its own $\chi^2 = N$. Its control runs the same solver without the pairing: each model soft-thresholded on its own.

Coupling	Kind	What it assumes	Iterations	EC2 machine	Time (min)
No coupling	—	Nothing: the two models only share the mesh.	44	c5.9xlarge	22
Cross-gradient	structural	The gradients of the two models are parallel, or one of them is zero (Gallardo & Meju 2003).	45	c5.9xlarge	26
Joint total variation	structural	The two models change in the same places (Haber & Holtzman Gazit 2013).	44	c5.9xlarge	28
Linear correspondence	petrophysical	One linear relation in every cell: density contrast = 0.5 $\times$ susceptibility, the ratio of the two upper bounds.	52	c5.9xlarge	26
Group lasso	joint sparsity	Few cells are anomalous, and a cell is anomalous in both models or in neither (Utsugi 2025).	8 values of λ_1	c5.18xlarge	124
L1 + L2 by ADMM, no coupling (the group lasso's control)	control	Few cells are anomalous in each model, no pairing: what the group lasso's solver does without the coupling.	8 values of λ_1	c5.18xlarge	66

The couplings are those of `geoinv3d/methods/coupling.py` (docs/joint_couplings.md); the group lasso of `geoinv3d/methods/group_lasso.py` (docs/group_lasso_joint.md).

2 Results

2.1 Data fit

Coupling	Gravity χ^2/N	RMS (mGal)	Max residual (mGal)	Magnetic χ^2/N	RMS (nT)	Max residual (nT)
Single inversions (same settings)	0.93			0.90		
No coupling	0.95	0.49	4.9	0.95	47	585
Cross-gradient	1.10	0.53	4.9	0.88	53	912
Joint total variation	1.09	0.52	5.1	1.03	53	729
Linear correspondence	1.04	0.51	4.7	1.04	42	568
Group lasso	1.04	0.51	5.8	0.97	55	844
L1 + L2 by ADMM, no coupling (the group lasso's control)	1.05	0.51	5.7	0.96	55	815

Red: χ^2/N outside 0.7–1.3. The first row gives the single inversions with the same settings.

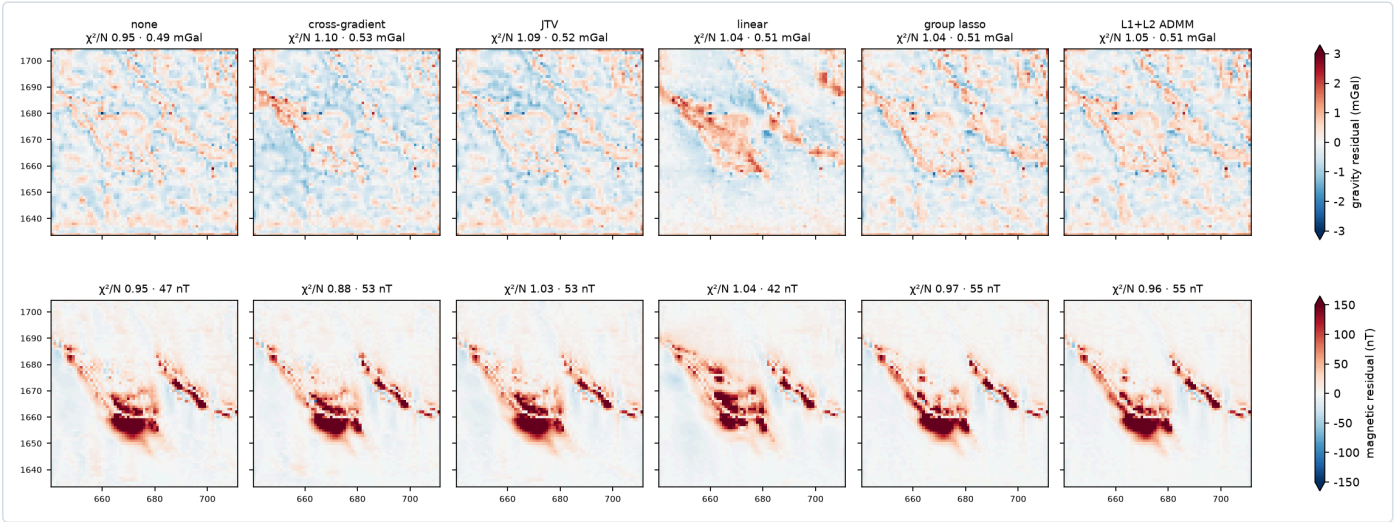


Figure 1 Residuals (observed – predicted) of the gravity data (top, ± 3 mGal) and the magnetic data (bottom, ± 150 nT) for each coupling.

Every run fits both datasets to their errors: χ^2/N lies between 0.88 and 1.10 for gravity and magnetics alike. The data fit therefore does not rank the couplings; what they do differently shows in the models (Sections 2.2 and 2.3).

Inverted together without a coupling, the two models are those of the single inversions (cell-by-cell correlation 0.95 for density and 0.76 for susceptibility): the joint set-up itself changes nothing. The positive magnetic residual south of the Sandur belt is there in every run: no coupling repairs what a magnetization along the present field cannot explain (Section 3).

2.2 What each coupling does to the models

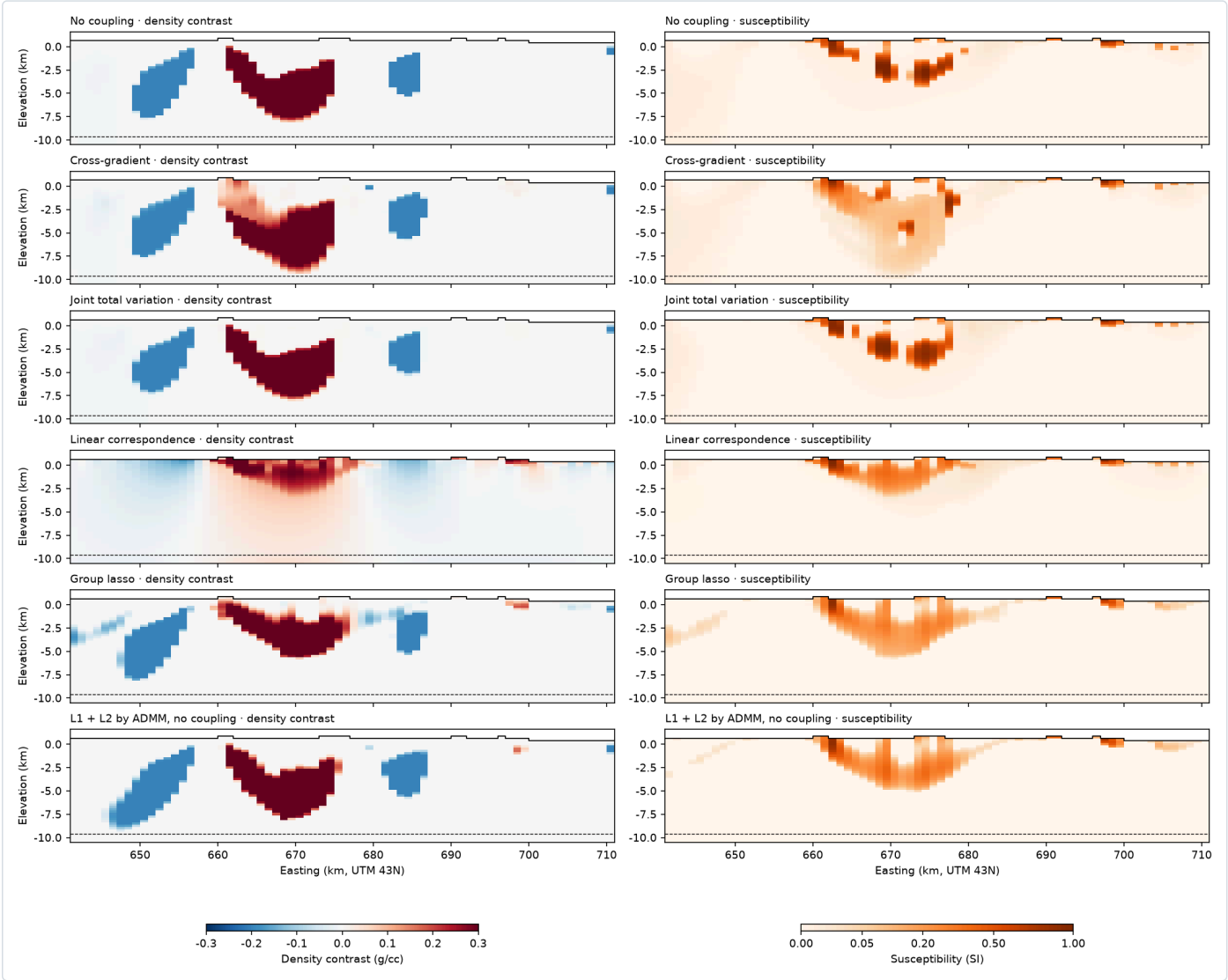


Figure 2 E–W sections through the Sandur belt (northing 1667.5 km): density contrast (left, ± 0.3 g/cc) and susceptibility (right, square-root scale to 1 SI) for each coupling. The black line is the ground of the mesh, the dashed line the base of the core.

Coupling	Density: core	below the core	main high, half-max km	centroid km	correlation with the uncoupled model	Susceptibility: core	below the core	Sandur belt, centroid km	SI·km³	correlation with the uncoupled model
No coupling	89%	9%	3.0–7.5	5.3	1.00	73%	7%	2.0	150	1.00
Cross-gradient	90%	9%	2.5–8.5	5.6	0.92	70%	8%	2.7	165	0.60
Joint total variation	88%	10%	3.0–7.2	5.2	0.97	73%	10%	2.1	147	0.61
Linear correspondence	41%	45%	0.0–3.8	4.9	0.37	55%	19%	1.4	153	0.43
Group lasso	89%	3%	2.0–5.2	3.7	0.74	75%	6%	2.5	157	0.58
L1 + L2 by ADMM, no coupling (the group lasso's control)	92%	6%	3.2–7.5	5.4	0.92	74%	6%	2.3	152	0.61

Depths below the ground. Main high: the density profile under the main Bouguer high (671.5, 1664.5 km). Sandur belt: susceptibility $\times$ volume in the box of the magnetic report. Correlations are cell by cell in the core columns. Red: less than half in the core, or more than 30% below it.

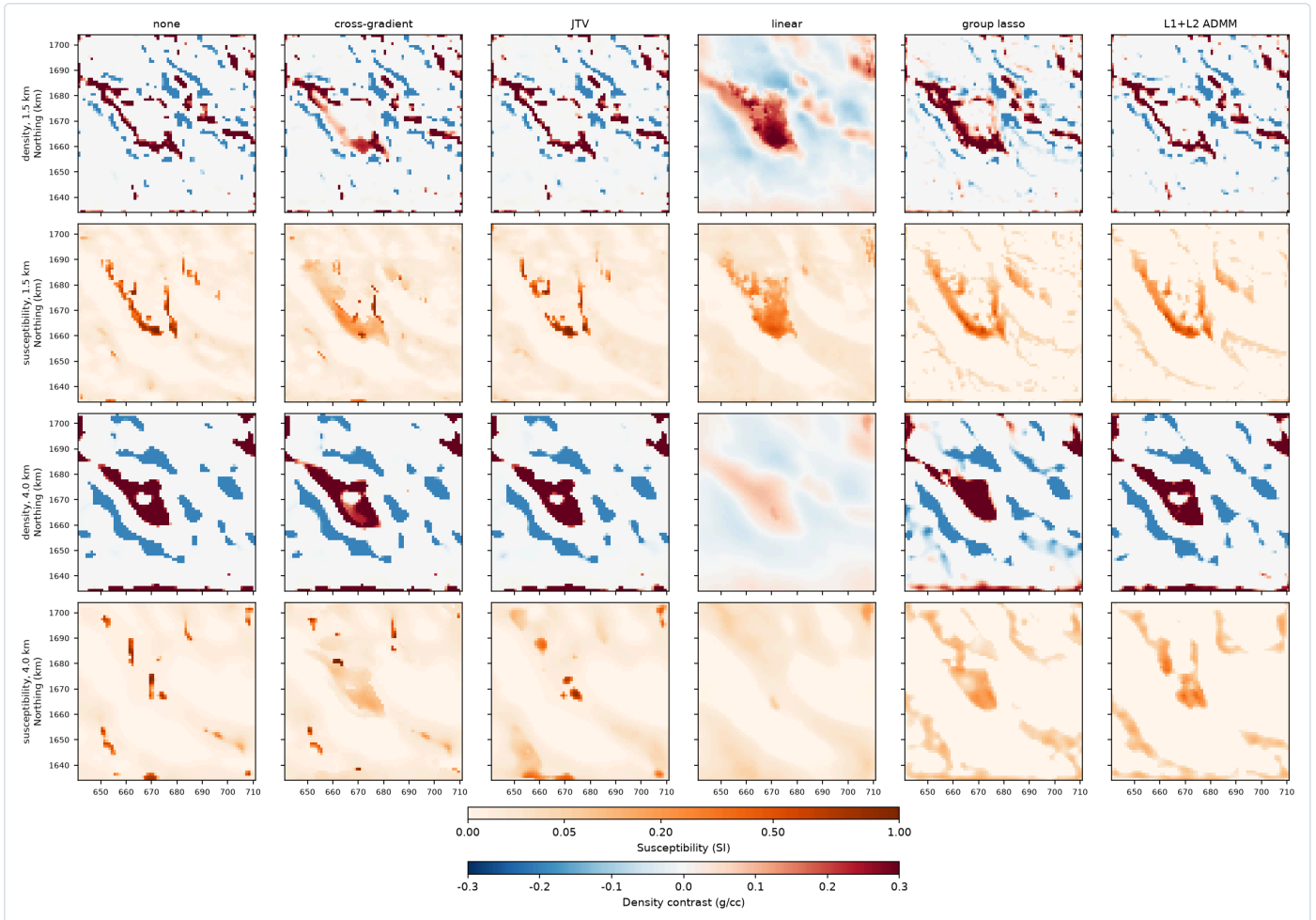


Figure 3 Layers at about 1.5 km and 4 km below the mean ground: density contrast and susceptibility for each coupling (core area only).

- **Cross-gradient.** It does what it is built for: the cross-gradient measure of Section 2.3 falls from 0.44 to 0.07, with both datasets fitted (χ^2/N 1.10 and 0.88). It gets there mostly by the second way of making a cross product zero: only 2% of the susceptibility model's edges lie on an edge of the density model (32% uncoupled), so where one model has an edge the other is flat. The density model hardly changes (correlation 0.92 with the uncoupled one); the susceptibility model changes more (0.60), and its centroid in the belt moves from 2.0 to 2.7 km.
- **Joint total variation.** It rewards edges in the same cells, and the share of susceptibility edges on a density edge rises from 32% to 47%; the directions of the gradients are not constrained (measure 0.39). 38% of the magnetic cells are anomalous in density (21% uncoupled). The density model stays as it was (0.97); both datasets are fitted (χ^2/N 1.09 and 1.03) in 44 iterations.
- **Linear correspondence.** The two models become one: 93% of the magnetic cells are anomalous in density too, and $|\text{density}|$ correlates with susceptibility at 0.71. The relation does not describe this belt's rocks (Section 4.3), and the density model pays for it: the dense body under the main high has its half-maximum at 0.0–3.8 km, against 3.0–7.5 km uncoupled, and what the relation cannot place is put below the core (45% of the density model, against 9%). Both datasets are still fitted, to χ^2/N 1.04 and 1.04: the fit does not warn.
- **Group lasso.** Both datasets are fitted (χ^2/N 1.04 and 0.97), after 1 round of reweighting the datasets (weights 0.62 for gravity and 1.60 for magnetics: along the λ , sweep the gravity data are fitted sooner), and the models stay where the data resolve them (89% of the density and 75% of the susceptibility model in the core). The pairing shows: 81% of the magnetic cells are anomalous in density, against 30% in its control without the pairing and 21% in the uncoupled SimPEG run, and $|\text{density}|$ correlates with susceptibility at 0.52 (0.06 uncoupled). It also makes the two structures alike: 75% of the susceptibility edges lie on a density edge (32% uncoupled), more than with any coupling but the linear correspondence, and the cross-gradient measure falls to 0.13 (0.44 uncoupled; 0.07 for the cross-gradient itself). The pairing reshapes the density model too (correlation 0.74 with the uncoupled one): the dense body under the main high rises to a half-maximum at 2.0–5.2 km (uncoupled 3.0–7.5 km; its control 3.2–7.5 km; centroid 3.7 km against 5.3 km), towards the magnetic rock, and 3% of the density model lies below the core (9% uncoupled). The susceptibility model spreads through the dense body (correlation 0.58 with the uncoupled one; centroid in the belt 2.5 km, against 2.0 km).
- **The group lasso's control** (the same solver, weighting and balance, each model on its own) fits the data to χ^2/N 1.05 and 0.96 and keeps 92% and 74% of its models in the core; 30% of its magnetic cells are anomalous in density. What the group lasso adds to it is the pairing of the two supports.

2.3 How alike the two models are

Measured in the core of the mesh. The cross-gradient measure is the sum of $|\nabla \rho \times \nabla \chi|^2$ over the sum of $|\nabla \rho|^2 |\nabla \chi|^2$: 0 when the gradients are parallel wherever both exist, 1 when they are perpendicular. An edge is a cell whose squared gradient exceeds 1% of the model's largest; a cell is anomalous when its value exceeds a tenth of the model's largest.

Coupling	Cross-gradient (0 = parallel gradients)	Susceptibility edges on a density edge	Magnetic cells that are dense or light	Cell-by-cell correlation of density and susceptibility	Correlation of the integrated maps
No coupling	0.44	32%	21%	0.06	0.20
Cross-gradient	0.07	2%	31%	0.06	0.26
Joint total variation	0.39	47%	38%	0.13	0.32
Linear correspondence	0.03	90%	93%	0.71	0.59
Group lasso	0.13	75%	81%	0.52	0.39
L1 + L2 by ADMM, no coupling (the group lasso's control)	0.37	39%	30%	0.13	0.33

Green: clearly more alike than the uncoupled models (the first row).

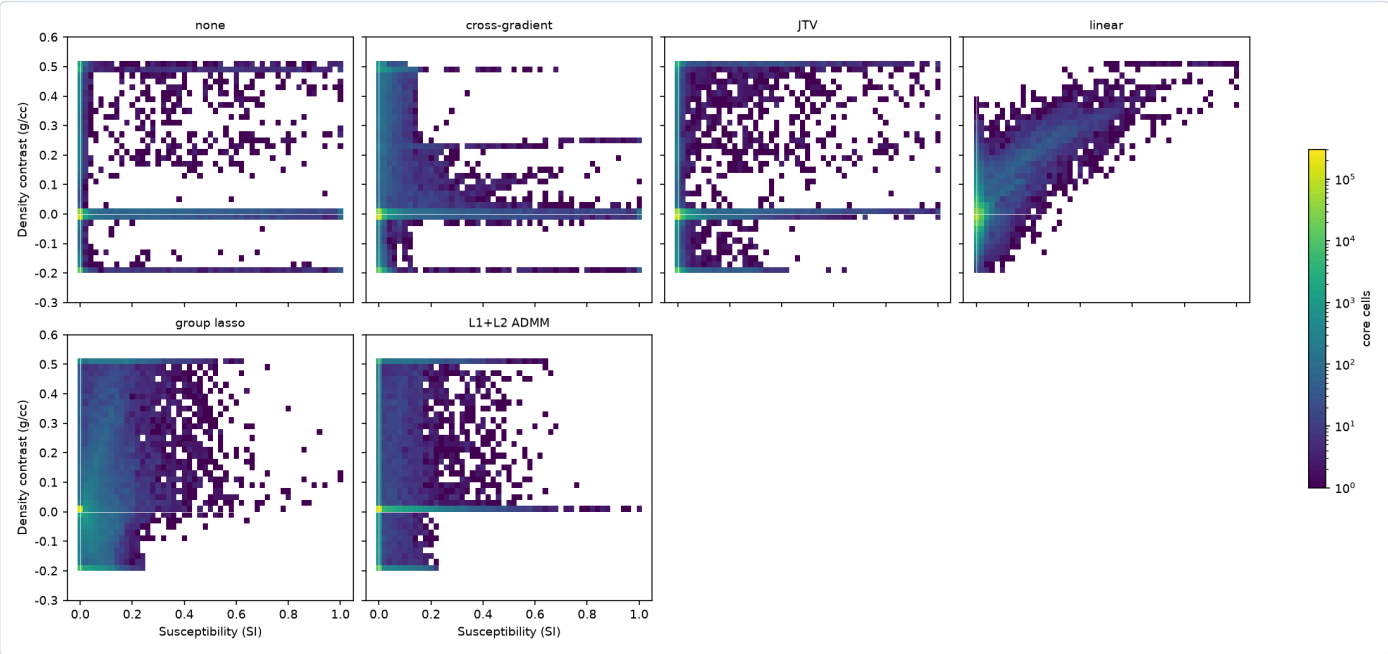


Figure 4 Density contrast against susceptibility, cell by cell in the core (logarithmic counts). Each coupling leaves its own signature.

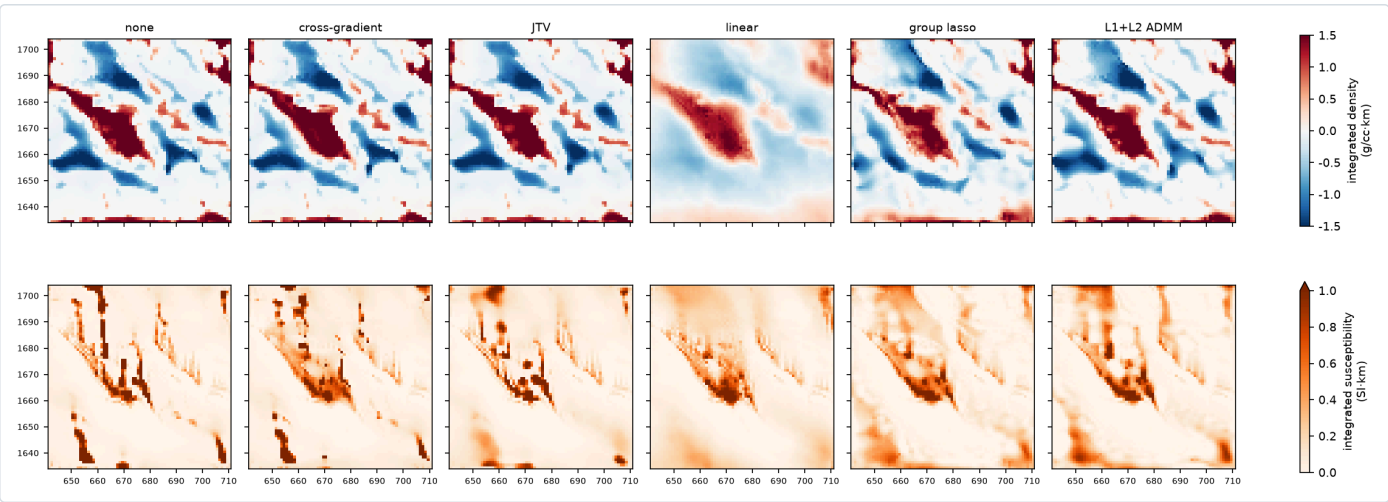


Figure 5 Vertically integrated density (top) and susceptibility (bottom) for each coupling.

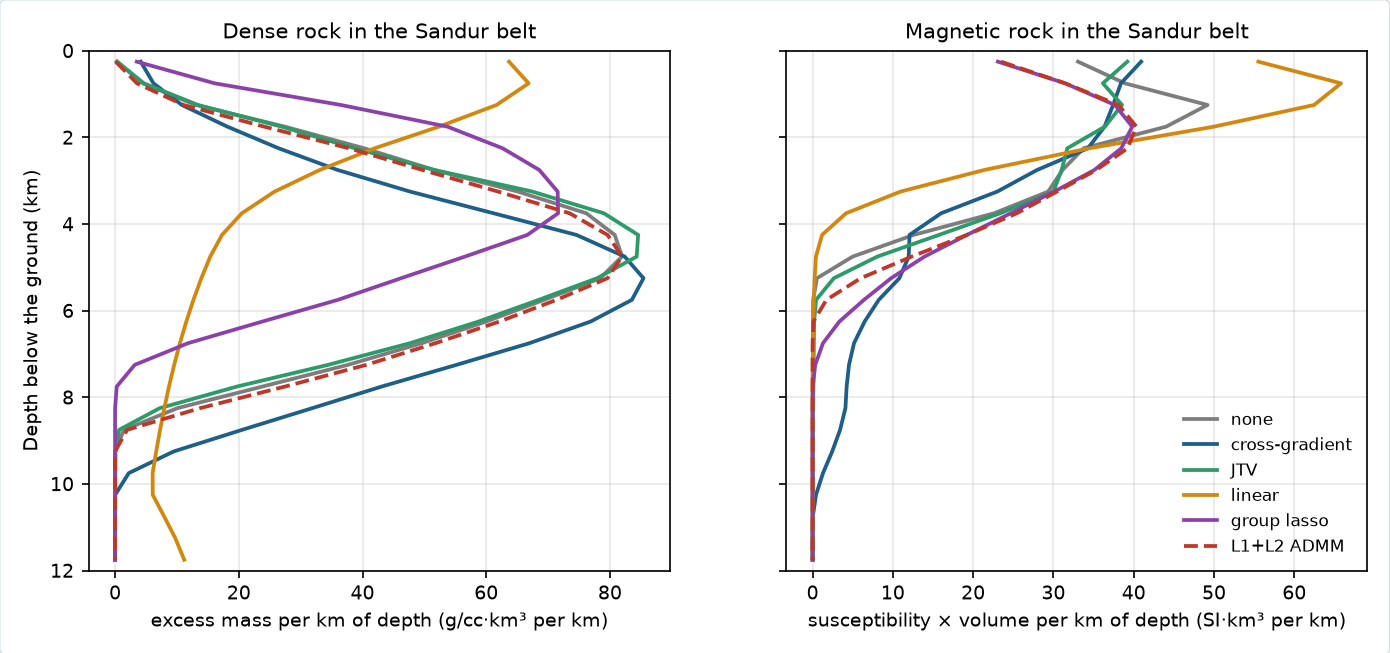


Figure 6 Dense rock (left: positive density contrast × volume) and magnetic rock (right: susceptibility × volume) in the Sandur belt, per kilometre of depth below the ground, for each coupling.

3 The magnetic residual: remanent magnetization

A susceptibility model magnetizes every cell along the present field, and a body so magnetized gives a fixed shape to its anomaly: the ratio of its high to its low, and where they lie. The positive anomaly on the southern side of the Sandur belt is fitted worse than the rest: the single magnetic inversion underfits 92 stations by more than 150 nT (easting 664–707 km, northing 1655–1673 km; RMS 321 nT there), and every joint run leaves 88–107 such stations, with an RMS of 224–360 nT at the same 92.

The magnetic data were therefore inverted once more with a magnetization vector in every cell (SimPEG's magnetization-vector inversion: three components of effective susceptibility per cell, a sparse regularization of their amplitude with the same norms, depth weighting and bounds, $|m_i| \leq 1$ SI), which allows a magnetization in any direction. It fits the data to χ^2/N 1.05 with an RMS of 28 nT, against 49 nT for the susceptibility model, and its largest residual is 309 nT against 680 nT. At the 92 underfitted stations the RMS residual falls from 321 to 129 nT (mean |residual| 293 → 105 nT); 22 of them are still off by more than 150 nT (16 above, 6 below the model), so the vector model explains most, not all, of the anomaly.

The strongly magnetized cells (the 1,203 above a tenth of the largest amplitude) point at an inclination of 74° and a declination of -79° (their resultant; coherence 0.75, where 1 means all parallel), far from the present field (I 19°, D -1°). The data ask for a magnetization that is not along the present field: remanent magnetization, common in banded iron formation. The amplitude model is compact: 79% of it lies in the core (71% of the susceptibility model).

For the joint inversions this means that the susceptibility models of the belt are an approximation where the rock is remanent: a coupling can move the susceptibility towards the density model, but it cannot give it the direction the data ask for. The couplings of this report act on the induced susceptibility.

Magnetic inversion ($\beta = 1$)	χ^2/N	RMS (nT)	Max residual (nT)	In the core	Below the core	Sandur belt, centroid km	EC2 machine	Time (min)
Susceptibility along the present field	0.90	49	680	71%	10%	2.1	c5.4xlarge	14
Magnetization vector (MVI)	1.05	28	309	79%	10%	2.1	c5.9xlarge	27

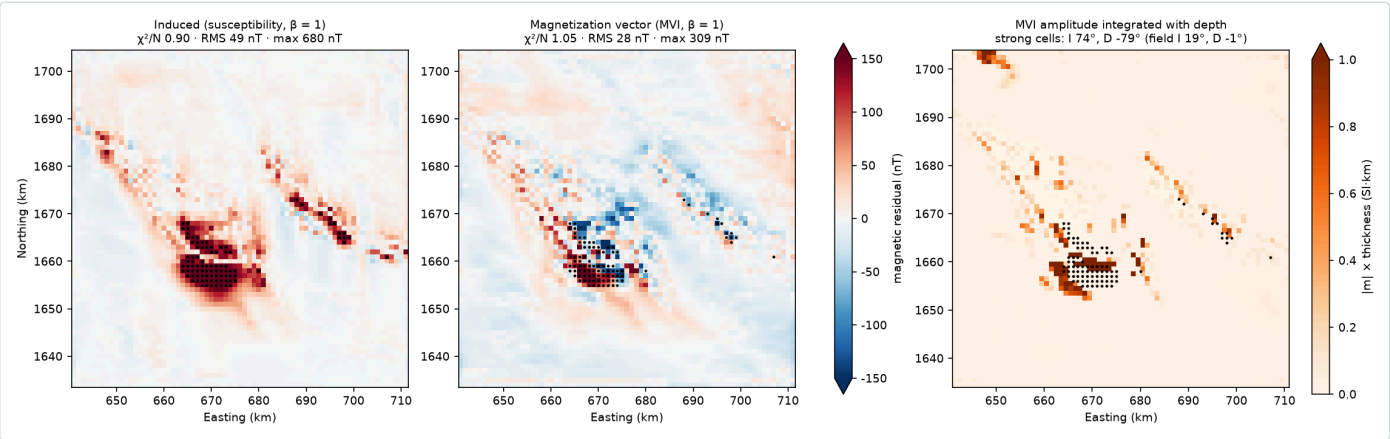


Figure 7 Magnetic residuals of the susceptibility inversion and of the magnetization-vector inversion (± 150 nT; black dots: the stations the susceptibility model underfits by more than 150 nT), and the amplitude of the magnetization vector integrated with depth.

4 Choosing a coupling

There is no known model of this area to compare the results with, and every coupling fits the data equally (Section 2.1), so the data do not rank the couplings. What this study can say falls into three kinds, of decreasing certainty: what each coupling does to the two models, which is a property of the method and would be the same whatever the true rock (4.1, 4.2); what the magnetic data themselves show, the remanence (Section 3); and which coupling suits this belt, a judgement from the geology and the rock samples, not from the inversions (4.3). Section 4.4 turns them into a first choice for other geological settings.

4.1 What each coupling does

Coupling	What it asks of the models	What it did here	Why
Cross-gradient	The gradients of the two models are parallel wherever both change: $\nabla\rho \times \nabla\chi = 0$.	Cross-gradient measure $0.44 \rightarrow 0.07$, but only 2% of the susceptibility edges on a density edge (32% uncoupled). Density hardly changes (0.92); susceptibility smoother (0.60), centroid in the belt $2.0 \rightarrow 2.7$ km.	A cross product is also zero where one of the two gradients is zero: keeping the edges of the two models apart is cheaper than lining them up.
Joint total variation	The two models change in the same places: one total variation of both, $\sqrt{(\nabla\rho ^2 + \nabla\chi ^2)}$.	Susceptibility edges on a density edge 32% $\rightarrow$ 47%; magnetic cells anomalous in density 21% $\rightarrow$ 38%. Density as uncoupled (0.97); fit 1.09 / 1.03.	$\sqrt{a^2 + b^2} < a + b$: two edges in the same cell cost less than two edges apart. Neither the values nor the direction of the gradients are tied.
Linear correspondence	One linear relation between the values in every cell: density = $0.5 \times$ susceptibility.	93% of the magnetic cells anomalous in density; the dense body under the main high $3.0\text{--}7.5 \rightarrow 0.0\text{--}3.8$ km; 45% of the density model below the core (9% uncoupled). Fit 1.04 / 1.04.	A cell can be dense only if it is magnetic: density goes where the magnetic data put the magnetic rock (near the surface) or where they cannot see it (below the core) (4.2).
Group lasso	Few cells are anomalous, and a cell is anomalous in both models or in neither; no ratio between the values.	81% of the magnetic cells anomalous in density (30% in its control, 21% uncoupled); 75% of the edges shared; the dense body $3.0\text{--}7.5 \rightarrow 2.0\text{--}5.2$ km; susceptibility spread through the dense body. Fit 1.04 / 0.97.	The group norm $\ (\rho, \chi)\ < \rho + \chi $: density is cheaper in a cell that already holds susceptibility; with no fixed ratio it can still stay elsewhere, so the shift is partial (4.2).

Measured in the core of the 1 km runs against the uncoupled run (Sections 2.2 and 2.3). The behaviour follows from what each coupling asks of the models; it describes the method, not whether its result is right.

4.2 Why the couplings that tie the values lift the dense body

The two datasets do not see depth alike. Per unit volume, the magnetic data lose sight of a cell much faster than the gravity data: at 4–8 km below the ground a cell counts 0.22 of what it counts at the surface for gravity but 0.10 for the magnetic data, and below the core 0.11 against 0.027 (the field of a magnetic dipole falls off one power of the distance faster than that of a mass). The magnetic data therefore fix where the magnetic rock is, near the surface, much more firmly than the gravity data fix the depth of the dense rock.

Linear correspondence. With density = $0.5 \times$ susceptibility in every cell, a cell can be dense only if it is magnetic too: the relation asks for 2 SI of susceptibility per g/cc of density, 0.6 SI for a contrast of 0.3 g/cc. The dense body at 3.0–7.5 km of the uncoupled run would need that much susceptibility at depth, whose magnetic anomaly the data do not have; so density is allowed only where the magnetic rock is — the shallow sheets, which carry the dense body up to 0.0–3.8 km under the main high — and where the magnetic data can no longer object: below the core, which receives 45% of the density model (9% uncoupled). The susceptibility model gets shallower too (centroid in the belt 1.4 km, against 2.0 km uncoupled).

Group lasso. The group norm of a cell, $\|(\rho, \chi)\|$, is less than $|\rho| + |\chi|$: density costs less in a cell that already holds susceptibility than in an empty one. So the dense rock is drawn towards the shallow magnetic cells, but, with no fixed ratio, it may still stay in non-magnetic cells at the full price: the dense body rises only part of the way, to 2.0–5.2 km, and little goes below the core (3%). Its control, the same solver without the pairing, keeps the dense body at 3.2–7.5 km: the lift comes from the pairing.

The couplings that leave the values free (the cross-gradient and the joint total variation) place no such condition on a cell, and the dense body stays where the gravity data and its own regularization put it. None of the depths is confirmed by the data — every run fits them equally — so the lift shows what the assumption does, not where the rock is.

Depth below the ground (km)	Gravity	Magnetics	Magnetics / gravity
0–2	1.00	1.000	1.00
2–4	0.43	0.295	0.68
4–8	0.22	0.099	0.45
8–10	0.15	0.048	0.33
below 10 (below the core)	0.11	0.027	0.26

Median column norm of the error-weighted sensitivities per km³ of cell, in the core columns of the 2 km mesh, relative to the cells 0–2 km below the ground.

4.3 Which coupling suits this belt

This is a judgement, not a result of the inversions: with no known model and every coupling fitting the data equally, the choice rests on what is known of the rocks. The single-method reports and the rock samples describe the Sandur belt as a thick body of dense, weakly magnetic rock (the greenstone pile; metabasalt and amphibolite samples of 2.94–2.98 g/cc and 0.0002–0.017 SI inside the block, Section 5.1) with thin sheets of strongly magnetic rock along its margins and its south-eastern closure (the banded iron formation), which are dense too (3.4 g/cc) but too thin to dominate the gravity. Against this, the assumptions of the couplings fare as follows.

- **One relation between the values (linear correspondence): contradicted by the samples.** Its premise, that the densest rock is the most magnetic, does not hold here: the greenstone is dense and hardly magnetic, the iron formation dense and strongly magnetic, two rocks with different relations. It can be set aside with some confidence, and its result shows the cost of the wrong premise: the dense body moved to 0.0–3.8 km and 45% of the density model below the core, with the data fitted as well as by any other coupling.
- **The same cells are anomalous (group lasso): true for the iron formation, not for the greenstone.** The iron formation is both dense and magnetic, the greenstone dense only. The group lasso does not force the greenstone to be magnetic, but it draws the magnetic rock into the dense cells (81% of the magnetic cells anomalous in density, against 21% uncoupled) and the dense body towards the magnetic rock (2.0–5.2 km). It fits where the magnetic rock is the dense rock; over the whole belt its premise holds only in part.
- **Same boundaries, free values (structural couplings): consistent with the geology.** The iron formation bounds the greenstone, so edges of the susceptibility model should lie on edges of the density model, while their values stay unrelated. The joint total variation asks for exactly that and delivers it in part (47% of the susceptibility edges on a density edge, 32% uncoupled); the cross-gradient asks only for parallel gradients and is satisfied, more cheaply, by keeping the edges apart (2%). Both leave the density model as the gravity data and its regularization make it (correlation 0.97 and 0.92 with the uncoupled one). Of the two, the joint total variation is the one whose result matches its premise here.
- **Remanence.** The strongest magnetic rock of the belt is not magnetized along the present field (Section 3). Every coupling here acts on the induced susceptibility, so in the south of the belt the coupled susceptibility models inherit an approximation that no coupling removes.

The judgement could be tested: on a synthetic model built from this geometry (thin magnetic sheets on the margins of a thick dense body), where the truth is known, the couplings can be compared directly; density and susceptibility logs, or measured samples at the scale of a cell, would turn the premises into constraints; oriented samples of the iron formation would tell whether the direction of Section 3 is its remanence.

4.4 Which coupling for which geology

A coupling is an assumption about how the two properties are related. The first choice follows from what is known of the rocks before the inversion; the last column says what to check in the result.

Geological setting	First choice	Also	What to check
One rock carries both anomalies (a body that is both dense and magnetic: a mafic–ultramafic intrusion, massive iron formation or magnetite ore alone)	Group lasso	PGI, if the rock's density and susceptibility are measured	Whether the paired support is geologically one body; the depth of the dense body moves towards the magnetic one
Different rocks share their boundaries (magnetic rock along the contacts or margins of dense rock, as the iron formation along the greenstone of this belt, or a magnetic unit along a fault that also bounds a dense one)	Joint total variation	Cross-gradient	The share of common edges: the cross-gradient can meet its condition by keeping the edges apart (2% here). Avoid the linear correspondence and the group lasso: they move one model to the other
A few rock types with known properties at the scale of a cell (logs, measured samples)	PGI, with the measured units	Joint total variation	Whether each unit's cells make geological sense; units chosen by us are assumptions, not constraints
One rock type with a known, fixed relation between density and susceptibility (fitted to samples of that rock)	Linear correspondence	Group lasso	The depth and the share below the core of the density model: a wrong relation moves it, and the data fit does not warn
The relation is not known	No coupling first: invert separately and compare the two models (common edges, shared support)	Then the coupling whose assumption the comparison supports	Report what the couplings agree on; where they differ, the data do not decide
The magnetic anomalies do not fit an induced magnetization (a systematic residual, highs and lows in the wrong ratio)	A magnetization-vector inversion of the magnetic data	—	Couplings of the induced susceptibility are an approximation where the rock is remanent

From the behaviour of Section 4.1, the belt of this report and the synthetic comparison of 30 September (three dense bodies, of high, low and no susceptibility), where the group lasso recovered the bodies best because each magnetic body was a dense body. PGI was not run in this series.

5 Constraints from the rock samples

Sections 2–4 bound the density contrast at –0.2 to +0.5 g/cc and weight depth with $\beta = 1$ in both models, as the single-method reports did. The rock samples of the area measure the density of each rock family. This section uses them as bounds and, since the belt crops out, tests them together with the depth weighting of the gravity model: for the uncoupled run and the joint total variation, first on the 2 km mesh and then at full resolution.

5.1 The measured densities

Rock family	Inside the block: median (range), n	Region: median (range), n	Contrast to 2.66 g/cc (block / region median)	Susceptibility inside the block: median (5–95%), 10 ^{–3} SI
Metabasalt / amphibolite	2.94 (2.81–2.99), n = 6	2.98 (2.64–3.33), n = 40	+0.28 / +0.31	7.6 (0.21–17)
Dolerite / gabbro	3.01 (2.96–3.03), n = 3	3.02 (2.77–3.24), n = 53	+0.35 / +0.36	0.38 (0.27–1.7)
Banded iron formation	3.39 (3.38–3.40), n = 2	3.38 (3.03–3.40), n = 3	+0.73 / +0.72	54 (53–55)
Granite	2.63 (2.60–2.74), n = 19	2.64 (2.52–3.10), n = 213	–0.03 / –0.02	1.1 (0–7.6)
Gneiss / TTG	2.62, n = 1	2.66 (2.59–3.02), n = 38	–0.04 / 0.00	0.66
Schist, phyllite, quartzite	2.69, n = 1	2.65 (2.62–2.69), n = 4	+0.03 / –0.01	0.04

Rock family	Inside the block: median (range), n	Region: median (range), n	Contrast to 2.66 g/cc (block / region median)	Susceptibility inside the block: median (5–95%), 10 ⁻³ SI
Felsic volcanic	– 2.64 (2.60–2.68), n = 2		–0.02	–

Rock samples of the area (toposheets 57A and 57B), grouped by rock family, from the density report of 30 September 2026 (`data/rock_properties/measured_by_rock_type.csv`). Block: the 70 × 70 km area of this report; region: the two toposheets. Densities in g/cc.

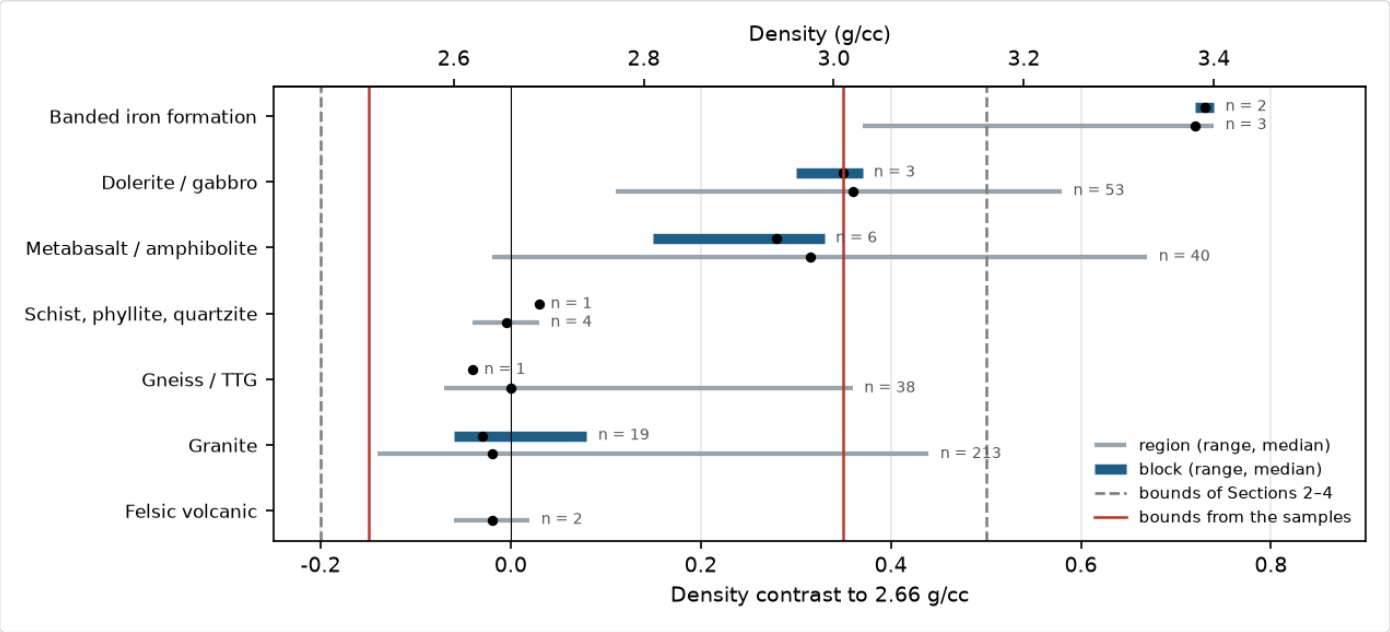


Figure 8 The measured densities as contrasts to the 2.66 g/cc background: range and median of each rock family inside the block (dark) and in the region (light), with the bounds of Sections 2–4 (grey, dashed) and those from the samples (red).

Against the background of 2.66 g/cc, the metabasalt and amphibolite of the belt are +0.28 (inside the block, median of 6) to +0.31 g/cc (region, 40), the dolerite and gabbro +0.35 to +0.36, and the granite and gneiss of the host –0.03 and 0.00; the banded iron formation, at 3.39 g/cc, is +0.73. The bounds of Sections 2–4 are wider than the belt's rocks: +0.5 g/cc lies above the median of every family but the iron formation, and –0.2 g/cc below the lightest sample of the region (granite, 2.52 g/cc, –0.14).

The bounds tested are a lower bound of –0.15 g/cc, just below that sample, and an upper bound of +0.30, +0.35 or +0.40 g/cc: the metabasalt with some gabbro, within the spread of the samples (about ±0.1 g/cc around their median). The iron formation exceeds all of them, but it lies in thin sheets on the margins of the belt that cannot carry the gravity high alone; where it lies, the density model can only reach the bound. The susceptibility bounds stay 0–1 SI: the samples of the iron formation (0.054 SI) are weathered, while the magnetic data ask for up to 1 SI at the scale of a cell (the magnetic report).

5.2 Bounds and depth weighting on the 2 km mesh

Coupling	Density bounds (g/cc)	β gravity / magnetics	Gravity χ²/N	Magnetic χ²/N	Dense cells at the upper bound	Main high, half-max km	centroid km	Column mass (g/cc·km)	Dense rock of the belt within 1 km of the ground	Magnetic rock within 1 km	Susceptibility edges on a density edge	Magnetic cells that are dense or light
No coupling	–0.20 / +0.50	1 / 1	0.90	0.90	22%	2.5–8.0	5.3	2.46	1%	31%	41%	26%
No coupling	–0.15 / +0.40	1 / 1	0.93	0.91	31%	2.0–8.0	5.1	2.25	1%	30%	41%	31%
No coupling	–0.15 / +0.35	1 / 1	1.03	0.98	38%	2.0–8.5	5.2	2.23	1%	31%	47%	32%
No coupling	–0.15 / +0.30	1 / 1	0.93	1.04	45%	1.5–9.0	5.2	2.21	2%	31%	49%	37%
No coupling	–0.15 / +0.35	0.5 / 1	0.90	0.84	21%	0.5–5.5	2.9	1.62	10%	30%	42%	35%
No coupling	–0.15 / +0.35	0.5 / 0.5	0.93	0.84	21%	0.5–5.5	3.1	1.62	10%	48%	60%	49%
Joint total variation	–0.20 / +0.50	1 / 1	0.90	0.91	18%	3.0–7.5	5.1	2.24	1%	42%	62%	38%
Joint total variation	–0.15 / +0.40	1 / 1	0.97	0.97	28%	2.0–8.0	5.1	2.22	1%	43%	66%	44%
Joint total variation	–0.15 / +0.35	1 / 1	1.12	1.07	41%	2.0–8.5	5.2	2.27	2%	39%	67%	40%
Joint total variation	–0.15 / +0.30	1 / 1	1.05	1.14	38%	2.0–9.0	5.6	2.33	2%	34%	77%	61%
Joint total variation	–0.15 / +0.35	0.5 / 1	0.93	0.94	19%	0.0–5.0	2.7	1.57	11%	48%	83%	60%

Coupling	Density bounds (g/cc)	β gravity / magnetics	Gravity χ^2/N	Magnetic χ^2/N	Dense cells at the upper bound	Main high, half-max km	centroid km	Column mass (g/cc·km)	Dense rock of the belt within 1 km of the ground	Magnetic rock within 1 km	Susceptibility edges on a density edge	Magnetic cells that are dense or light
Joint total variation	-0.15 / +0.35	0.5 / 0.5	0.91	0.89	20%	0.5–5.0	2.7	1.55	11%	68%	84%	61%

Depths below the ground. Main high: the density profile under the main Bouguer high; column mass: its positive density $\times$ thickness. Within 1 km: the share of the positive density $\times$ volume (or of the susceptibility $\times$ volume) in the Sandur belt that lies within 1 km of the ground. Red: χ^2/N outside 0.7–1.3.

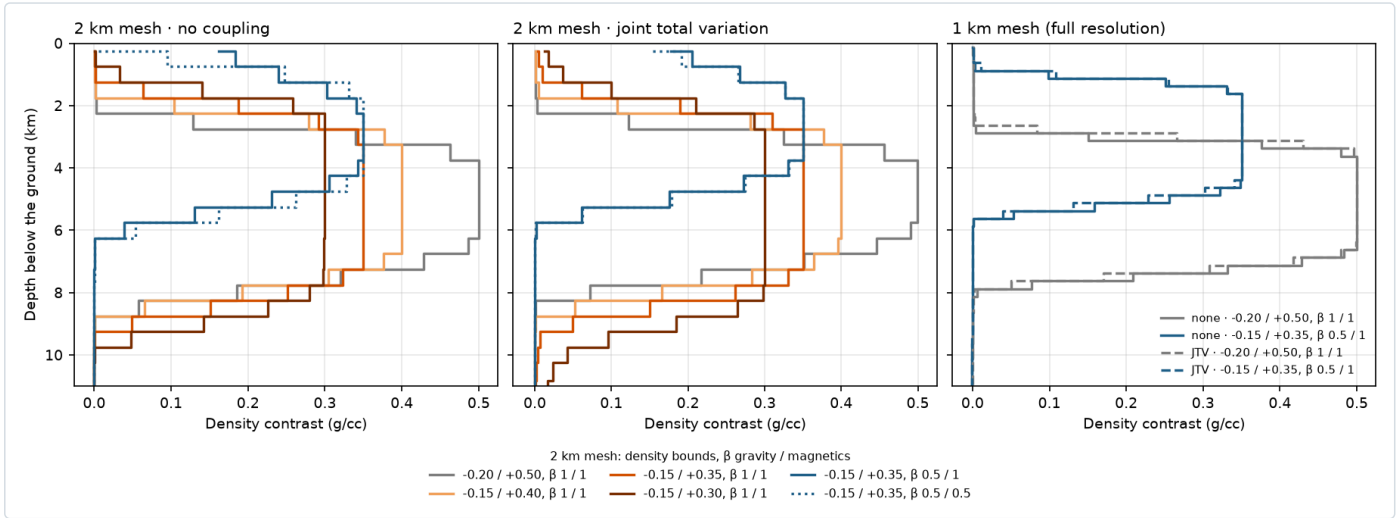


Figure 9 The density contrast under the main Bouguer high: on the 2 km mesh for each upper bound and depth weighting, uncoupled (left) and with the joint total variation (middle); at full resolution without and with the constraints (right).

- **Every combination fits both datasets** (χ^2/N 0.84–1.14): the data do not choose the bounds or the depth weighting.
- **The bounds stretch the dense body; they do not lift it.** With the upper bound lowered from +0.5 to +0.3 g/cc, about the same mass under the main high (2.21–2.46 g/cc·km) spreads over a taller column: uncoupled, the half-maximum depth range grows from 2.5–8.0 to 1.5–9.0 km, while the centroid stays at 5.1–5.6 km and at most 2% of the dense rock of the belt lies within 1 km of the ground.
- **The depth weighting sets where the body starts.** With $\beta = 0.5$ for the gravity model the dense body starts at the ground (0.5–5.5 and 0.0–5.0 km under the main high, uncoupled and with the joint total variation), 10–11% of the dense rock of the belt lies within 1 km of the ground, and the centroid rises to 2.7–2.9 km. The belt crops out, so this is the depth weighting the geology asks for; the data fit it as well.
- **Gravity fixes the mass of a column only for a body wide against its depth.** The column mass under the main high falls from 2.21–2.46 to 1.57–1.62 g/cc·km when the body moves up: a shallower body needs less mass under the high for the same anomaly. The thickness that mass $\div$ contrast implies therefore depends on the depth weighting as well as on the contrast.
- **The magnetic model keeps $\beta = 1$.** With $\beta = 0.5$ for both models, the share of the belt's magnetic rock within 1 km of the ground rises from 30% to 48% (uncoupled), and the root of the steep magnetic body at easting 671 km, which reaches 4 km with $\beta = 1$, is cut off. With $\beta = 1$ the magnetic model already reaches the ground, and the iron formation dips at 50–90° (the magnetic report): nothing asks for a shallower magnetic model.

5.3 Full resolution

Coupling	Density bounds (g/cc)	β gravity / magnetics	Gravity χ^2/N	RMS (mGal)	Magnetic χ^2/N	RMS (nT)	Main high, half-max km	centroid km	Dense rock of the belt within 1 km of the ground	Susceptibility below the core	Belt volume above half the largest susceptibility (km³)	Susceptibility edges on a density edge	Magnetic cells that are dense or light	Time (min)
No coupling	-0.20 / +0.50	1 / 1	0.95	0.49	0.95	47	3.0–7.5	5.3	1%	7%	140	32%	21%	22
No coupling	-0.15 / +0.35	0.5 / 1	0.91	0.48	0.94	48	1.0–5.2	3.2	7%	7%	141	31%	32%	23
Joint total variation	-0.20 / +0.50	1 / 1	1.09	0.52	1.03	53	3.0–7.2	5.2	1%	10%	131	47%	38%	28
Joint total variation	-0.15 / +0.35	0.5 / 1	0.92	0.48	0.91	50	1.0–5.2	3.1	7%	15%	83	56%	53%	28

As the table of Section 5.2; susceptibility below the core: its share of the susceptibility $\times$ volume (red above 30%). The rows with -0.20 / +0.50 are the runs of Sections 2–4.

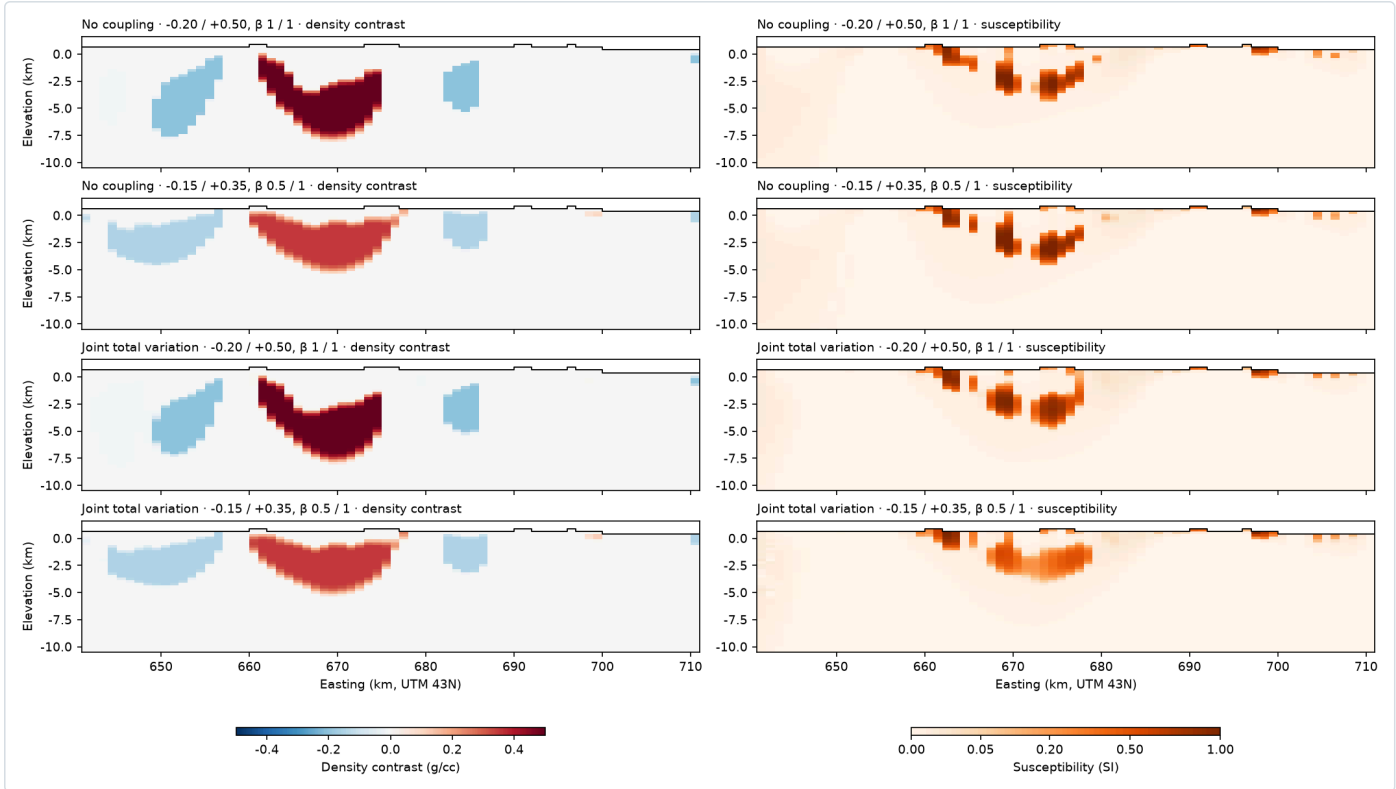


Figure 10 E–W sections through the Sandur belt (northing 1667.5 km) without and with the constraints: density contrast (left, ± 0.5 g/cc) and susceptibility (right, square-root scale to 1 SI).

- **The data are fitted as before:** χ^2/N 0.91–0.94, RMS 0.48 mGal and 48–50 nT (without the constraints 0.49–0.52 mGal and 47–53 nT).
- **The belt reaches the ground.** On the section through the belt the dense body meets the ground from easting 660 to 677 km; under the main high its half-maximum lies at 1.0–5.2 km (without the constraints 3.0–7.5 km; centroid 3.2 against 5.3 km), and 7% of the dense rock of the belt lies within 1 km of the ground (1%). The light bodies on either side now lie within about 4.5 km of the ground (down to about 7.5 km before).
- **The constraints change the density model more than any coupling.** Cell by cell, the constrained and unconstrained density models correlate at 0.42 (uncoupled) and 0.44 (joint total variation); within either setting the joint total variation and the uncoupled run correlate at 0.97 and 0.98. The uncoupled susceptibility model hardly changes (0.88).
- **With both models near the surface, the joint total variation lines them up more, and spreads the magnetic model.** It puts 56% of the susceptibility edges on a density edge (47% without the constraints) and 53% of the magnetic cells in dense or light ones (38%). But the two steep magnetic bodies at easting 668–678 km, up to 1 SI in the uncoupled model, become one wider body of about 0.2–0.5 SI that follows the dense keel: the volume of the belt above half the largest susceptibility falls from 141 km³ (uncoupled) to 83 km³, and 15% of the susceptibility model goes below the core (7% uncoupled). The geology describes steep sheets of iron formation; the data fit both.

5.4 What the constraints settle and what they do not

- **The choice rests on the rocks, not on the fit.** Every combination of bounds and depth weighting fits the data equally (χ^2/N 0.84–1.14 on the 2 km mesh); the bounds come from the samples and the depth weighting from the outcrop of the belt.
- **The depth of the base is not an independent result.** With these settings the half-maximum of the dense body under the main high ends at about 5.2 km, and the body (denser than +0.05 g/cc) at 5.5 km. A joint gravity–magnetic model of the belt in a conference abstract (Maurya, Ganguli & Rani 2023) describes a basin-like structure about 6 km deep; the two agree, but the abstract gives neither its profiles nor its density contrast, and its depth rests on its own assumed contrast.
- **At the bound.** 43–44% of the dense cells sit at +0.35 g/cc: the sparse norm ($p = 0$) makes blocky models, so the bound reads as the belt's mean contrast, not as the density of a cell.
- **The iron formation (+0.73 g/cc) exceeds the bound.** A higher bound only where the magnetic model places iron formation would need bounds per cell in the joint inversion.
- **The samples** are few inside the block for some rocks (6 metabasic, 2 of iron formation), located to 0.01° (about 1 km), from a survey not yet identified, and all from the surface, where weathered rock may read light.

6 The 2 km study

Every coupling was also run locally on a 2 km $\times$ 2 km $\times$ 500 m mesh (1,296 + 1,296 data):

Run (2 km mesh, local)	Gravity χ^2/N	Magnetic χ^2/N	Density range (g/cc)	Susceptibility range (SI)	Density in the core	Susceptibility in the core	Time (min)
No coupling	0.90	0.90	–0.20 to 0.50	0.00 to 1.00	88%	62%	0.7

Run (2 km mesh, local)	Gravity χ^2/N	Magnetic χ^2/N	Density range (g/cc)	Susceptibility range (SI)	Density in the core	Susceptibility in the core	Time (min)
Cross-gradient	0.93	0.80	-0.20 to 0.50	0.00 to 1.00	89%	57%	0.8
Joint total variation	0.90	0.91	-0.20 to 0.50	0.00 to 1.00	88%	69%	0.9
Linear correspondence	1.04	0.90	-0.20 to 0.50	0.00 to 0.98	54%	50%	0.8
Group lasso	0.96	1.02	-0.20 to 0.48	0.00 to 0.67	82%	62%	2.5
L1 + L2 by ADMM, no coupling (the group lasso's control)	0.97	1.02	-0.20 to 0.50	0.00 to 0.66	87%	61%	1.0

Red: χ^2/N outside 0.7–1.3.

- **The coarse runs behave like the full-resolution ones:** the cross-gradient lowers the cross-gradient measure, the joint total variation raises the share of common edges, the linear correspondence ties the values, and the group lasso pairs the supports. Each coupling took at most 2.5 minutes on a laptop (group lasso): enough to choose a coupling before paying for the full runs.
- **The group lasso on the 2 km mesh** fits the data to χ^2/N 0.96 and 1.02 and keeps 82% and 62% of its models in the core (89% and 75% at full resolution); 74% of its magnetic cells are anomalous in density (81% at full resolution).

7 Conclusions and recommendations

- **Certain: what each coupling does** (Section 4.1). The cross-gradient keeps the edges of the two models apart, the joint total variation lines them up, the linear correspondence and the group lasso move the dense body towards the magnetic rock. This is the behaviour of the methods on these data; it does not depend on what the rock really is.
- **Supported by the data: remanence** in the south of the Sandur belt (Section 3). The magnetic data of the belt should be inverted for a magnetization vector; a joint inversion that couples its amplitude, rather than the susceptibility, would carry the remanence into the joint model.
- **A judgement: for this belt, the joint total variation** (or the cross-gradient), read together with the uncoupled models (Section 4.3). Report what the couplings agree on; do not read the coupled susceptibility model as better resolved.
- **Use the rock samples as bounds and let the belt reach the ground** (Section 5): density -0.15 to +0.35 g/cc and $\beta = 0.5$ for the gravity model, $\beta = 1$ for the magnetic model. Read the density from the uncoupled run or the joint total variation (they agree), and the susceptibility from the uncoupled run or the magnetization-vector inversion: with the constraints, the joint total variation merges the steep magnetic sheets into one wider body. The couplings of Sections 2–4 were compared without the constraints; to compare them under the constraints, rerun them with these settings.
- **Elsewhere, choose the coupling from what is known of the rocks** (Section 4.4): the group lasso where the magnetic bodies are the dense ones, a structural coupling where different rocks share boundaries, the linear correspondence only for one rock type with a known relation, and no coupling first when the relation is unknown.
- **To test the judgement**, run the couplings on a synthetic model of this belt's geometry, where the truth is known, and constrain the premises with logs or measured samples; oriented samples of the iron formation for the direction of its magnetization.
- **Depth needs independent information.** The couplings do not supply it, and with the constraints it follows the bounds and the depth weighting; the mapped dips of the iron formation, boreholes, or the full model behind Maurya et al. (2023) would test it.

Files

- Parameters of every run: `examples/output/karnataka_joint/scripts/joint_params.py` (joint) and `deploy/ec2_multi_run.py` (the magnetic runs `beta1` and `beta1_mvi`); launched with `deploy/ec2_multi_run.py` and `scripts/run_lowres.py --out data/lowres_runs_fixed`
- This report and its figures: `scripts/make_figures.py --set v2` (with `scripts/constraint_figures.py`), `scripts/build_report_v2.py`, `scripts/report_text_v2.py`; results in `data/ec2_runs_fixed/`, `data/lowres_runs_fixed/` and `karnataka_magnetic/data/ec2_runs/`
- The constraints (Section 5): the variants of `joint_params.VARIANTS`, run with `scripts/run_lowres.py --bounds KEY` (2 km, `data/lowres_bounds/`) and `deploy/ec2_multi_run.py karnataka-joint --only none_rho35_gb05,joint_total_variation_rho35_gb05` (`data/ec2_runs_bounds/`); compared by `scripts/compare_bounds.py [--full]`; the rock samples in `data/rock_properties/`
- The single-method reports: `examples/output/karnataka_gravity_terrain/`, `examples/output/karnataka_magnetic/`